Smallest multipliers with leading digit one

and a conjecture of Fröhlich (OEIS A190302)

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Abstract

Let $a(n)$ be the smallest $h \geq 1$ such that the decimal expansion of $n \cdot h$ begins with the digit 1 (OEIS A190302 [1]). We prove the exact formula: writing $10^d \leq n < 10^{d+1}$, one has $a(n) = 1$ when $n < 2 \cdot 10^d$, and $a(n) = \lceil 10^{d+1}/n \rceil$ otherwise; in the second case the value lies in $\{2, 3, 4, 5\}$. In particular $a(n) \leq 5$ for every $n \geq 1$, which settles the conjecture recorded on the entry by Felix Fröhlich (Jul 28 2018: "$a(n) < 6$ for all n", verified there up to $n = 10022141$). The same bound $a(n) \leq 5$ was asserted without proof by C. R. Greathouse IV (2011) in the comments of OEIS A187285 [2]; the anteriority audit is archived in the verification archive (search corpus and date evidenced in the deposited `VERIFICATIONS/anteriorite.md`). The theorem is verified in Lean 4 with mathlib [3]: the kernel-checked proof uses only the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`), no `native_decide` and no `sorry`.

1 Statement of results

For $m \geq 1$, "the decimal expansion of m starts with 1" means $m \in [10^k, 2 \cdot 10^k)$ for some $k \geq 0$ (writing $[x, y)$ for the half-open interval $\{t : x \leq t < y\}$) – equivalently, the most significant decimal digit of m equals 1. OEIS A190302 (J. Krizek, May 07 2011, offset 1) [1] defines

$$a(n) \;=\; \min\{\, h \geq 1 : \text{the decimal expansion of } n \cdot h \text{ starts with } 1 \,\},$$

with first values $1, 5, 4, 3, 2, 2, 2, 2, 2, 1, \ldots$ The entry records the companion sequence A187285(n) [2] – the least multiple of n beginning with 1 – through the quotient identity $a(n) = \text{A187285}(n)/n$, and the conjecture of Felix Fröhlich (Jul 28 2018): $a(n) < 6$ for all n, verified there up to $n = 10022141$.

Theorem 1 (exact formula)**.** *Let $n \geq 1$ and let $d \geq 0$ be the integer with $10^d \leq n < 10^{d+1}$. Then*

$$a(n) = \begin{cases} 1, & n < 2 \cdot 10^d, \\ \lceil 10^{d+1}/n \rceil, & 2 \cdot 10^d \leq n < 10^{d+1}. \end{cases}$$

Corollary 2 (Fröhlich's conjecture, OEIS A190302)**.** $1 \leq a(n) \leq 5$ *for every $n \geq 1$; in particular* $a(n) < 6$.

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Corollary 3 (the five classes)**.** *For $d \geq 0$ and $h \in \{2,3,4,5\}$, an integer n with $d+1$ digits satisfies $a(n) = h$ iff*

$$\frac{10^{d+1}}{h} \;\leq\; n \;<\; \frac{10^{d+1}}{h-1};$$

and $a(n) = 1$ iff $n < 2 \cdot 10^d$. Each of the five values is attained (e.g. $a(2 \cdot 10^d) = 5$), so the bound 5 is sharp.

Theorem 1 and Corollary 2 are formally verified in Lean 4 with mathlib (§3); Corollary 3 is a direct consequence of the formula.

Prior art. The bound $a(n) \leq 5$ was asserted *without proof* by Charles R. Greathouse IV (Mar 07 2011) in the comments of OEIS A187285 [2]: "*a(n) is in* $\{n, 2n, 3n, 4n, 5n\}$" (where $a(n)$ there denotes $\mathrm{A187285}(n) = n \cdot a(n)$); the bound itself is not claimed as new.

2 Proof of Theorem 1

Fix $n \geq 1$ and d with $10^d \leq n < 10^{d+1}$.

Case $n < 2 \cdot 10^d$ (leading digit 1). The decimal expansion of n itself starts with 1, so $h = 1$ is admissible and $a(n) = 1$.

Case $2 \cdot 10^d \leq n < 10^{d+1}$ (leading digit ≥ 2). Set $c = \lceil 10^{d+1}/n \rceil$.

Admissibility. By definition of the ceiling, $10^{d+1} \leq n \cdot c$; and $n \cdot c \leq 10^{d+1} + n - 1 < 2 \cdot 10^{d+1}$ since $n < 10^{d+1}$. Hence $n \cdot c \in [10^{d+1},\, 2 \cdot 10^{d+1})$: its decimal expansion starts with 1, and $c \geq 1$ (as $n \cdot c \geq 10^{d+1} > 0$), so c is admissible.

Minimality. Let $h \geq 1$ be admissible: $n \cdot h \in [10^k,\, 2 \cdot 10^k)$ for some k. If $k \leq d$ then, using $h \geq 1$,

$$n \cdot h \;<\; 2 \cdot 10^k \;\leq\; 2 \cdot 10^d \;\leq\; n \;\leq\; n \cdot h,$$

a contradiction. Hence $k \geq d+1$ and $n \cdot h \geq 10^{d+1}$, which gives $c = \lceil 10^{d+1}/n \rceil \leq h$.

Therefore $a(n) = c$. The bound follows: since $n \geq 2 \cdot 10^d$, $10^{d+1}/n \leq 10^{d+1}/(2 \cdot 10^d) = 5$, so $c \leq 5$; and $c \geq 2$ since $n \cdot 1 = n < 10^{d+1}$ is not admissible. □

Corollary 2 is immediate: in both cases the admissible set is nonempty, so the minimum exists, is ≥ 1, and is ≤ 5. Corollary 3 is the ceiling inversion $\lceil 10^{d+1}/n \rceil = h \iff 10^{d+1}/h \leq n < 10^{d+1}/(h-1)$; each interval contains integers, so all five values occur.

3 Formal verification

The statement is formalized in Lean 4 [3] (toolchain `leanprover/lean4:v4.35.0`, mathlib `v4.35.0`, pinned manifest) in a single self-contained file `A190302.lean` (260 lines, `import Mathlib` only). The definitions read

$$\texttt{startsWithOne}(m) :\Longleftrightarrow\; \exists\, k,\; 10^k \leq m < 2 \cdot 10^k,$$

$$\texttt{a190302}(n) := \mathrm{sInf}\,\{h : \mathbb{N} \mid 1 \leq h \wedge \texttt{startsWithOne}(n \cdot h)\},$$

where sInf on $\mathbb{N}$ returns 0 on the empty set (totalizing convention; only the case $n = 0$ is affected, outside the OEIS domain). The formalized theorem is the *strengthened* statement

$$\texttt{A190302}: \quad \forall\, n \geq 1, \quad 1 \leq \texttt{a190302}(n) \ \wedge\ \texttt{a190302}(n) < 6,$$

the lower clause being necessary for faithfulness: without it, a hypothetical empty set of admissible h would satisfy $a(n) = \mathrm{sInf}\,\varnothing = 0 < 6$ vacuously. `A190302_lt_six` is the verbatim corollary $a(n) < 6$; `a190302_eq_one` and `a190302_eq_ceilDiv` export the exact formula of Theorem 1 (with $d = \texttt{Nat.log}\,10\,n$), together with d-parametrized variants `_of_pow`.

Faithfulness of the transcription is documented two ways. Internally, `startsWithOne_iff_getLastD` proves the interval predicate equivalent to the literal OEIS reading "most significant digit of m equals 1" via `Nat.digits` (the digit list is little-endian; the leading digit is its last element), and the values printed on the OEIS entry are spot-checked by `decide` on the eleven examples $a(1)$, $a(2)$, $a(3)$, $a(4)$, $a(5)$, $a(7)$, $a(11)$, $a(12)$, $a(19)$, $a(20)$ and $a(21)$. Externally, an independent re-derivation of the formalization from the raw OEIS text was produced by an external oracle before the file was finalized (blind prompt, verbatim entry): it converged to the same predicate shape, the same sInf-based term and the same strengthened clause; the exchange is archived in the artifact (`VERIFICATIONS/`).

`lake env lean A190302.lean`: exit 0, zero `sorry`. The report of `#print axioms A190302` gives the three standard axioms only – `propext`, `Classical.choice`, `Quot.sound` – with **no** `native_decide` and no `sorryAx`: the whole statement, including the finite example checks (`decide`-evaluated, hence kernel-reduced), is verified inside the Lean kernel.

Artifact. This note, the Lean file (`A190302.lean`), the pinned build files and the verification transcripts are archived together on Zenodo [4] (DOI `10.5281/zenodo.23957596`); source repository `github.com/gaudinfrederic/oeis-A190302` [5].

4 Remarks

- The companion sequence is thus exactly $\mathrm{A187285}(n) = n$ when the leading digit of n is 1, and $n \cdot \lceil 10^{d+1}/n \rceil \in [10^{d+1},\, 2 \cdot 10^{d+1})$ otherwise – the least multiple of n beginning with 1, recovering Greathouse's comment $\mathrm{A187285}(n) \in \{n, 2n, 3n, 4n, 5n\}$.
- The proof is one line in spirit – the first multiple of n at or beyond 10^{d+1} must begin with 1 – but the OEIS domain is $n \geq 1$ with no asymptotic range, so the value of the note is exactness for every n and the kernel-checked statement, not a new bound.
- In base $b \geq 2$ the same argument gives: if the leading base-b digit of n is 1 then $a_b(n) = 1$, else $a_b(n) = \lceil b^{d+1}/n \rceil \leq \lceil b/2 \rceil$ (for instance ≤ 5 in base 10, ≤ 2 in base 4; in base 2 always $a_2(n) = 1$). Remark only – not part of the Lean artifact.
- Numerical status: the conjecture was verified by Fröhlich on the entry up to $n = 10022141$; the exact formula was additionally tested against brute force on $[1, 5000]$ (total agreement) – direction tool, not part of the proof.

References

[1] J. Krizek, OEIS A190302 (May 07 2011); conjecture comment F. Fröhlich, Jul 28 2018. *The On-Line Encyclopedia of Integer Sequences*, N. J. A. Sloane ed., `https://oeis.org/A190302`.

[2] OEIS A187285 (least multiple of n beginning with 1); comment C. R. Greathouse IV, Mar 07 2011: "a(n) is in $\{n, 2n, 3n, 4n, 5n\}$", `https://oeis.org/A187285`.

[3] Lean 4 + mathlib (`v4.35.0`): theorem `A190302`, single file `A190302.lean`; `lake env lean`: exit 0, 0 `sorry`, axioms `[propext, Classical.choice, Quot.sound]`.

[4] F. Gaudin, *Smallest multipliers with leading digit one and a conjecture of Fröhlich (OEIS A190302)* - Zenodo deposit of this note, the Lean file and the verification transcripts, `https://doi.org/10.5281/zenodo.23957596`.

[5] F. Gaudin, source repository *oeis-A190302*, `https://github.com/gaudinfrederic/oeis-A190302`.